

MAT 1013: Discrete Mathematics

Session 10 : Sets, Set Algebra, Cartesian Products and Cardinality

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Topics

- Sets and methods of describing sets
- Membership, subsets, equality of sets
- Universal set and empty set
- Cartesian products
- Set operations and Venn diagrams
- Set algebra identities
- Proving set identities using mutual inclusion
- Proving set identities using Venn diagrams
- Power sets, cardinality and infinity
- Inclusion–Exclusion Principle
- Connection between set algebra, logic and Boolean algebra

Learning Outcomes

By the end of this lesson, students should be able to:

- Define sets, elements, subsets and equality of sets.
- Use roster, descriptive and set-builder methods.
- Distinguish between membership $\in$ and subset $\subseteq$.
- Construct Cartesian products.
- Perform union, intersection, difference and complement operations.
- Use Venn diagrams to represent set operations.
- State and apply basic set algebra identities.
- Prove set identities using mutual inclusion.
- Verify set identities using Venn diagrams.
- Compute power sets and cardinalities of finite sets.
- Explain the basic idea of countable and uncountable infinity.
- Apply the Inclusion–Exclusion Principle.

1. What is a Set?

A **set** is a well-defined collection of objects. The objects in a set are called its **elements**.

Examples

Here are some examples of sets:

$$A = \{1, 2, 3, 4\}, \quad V = \{a, e, i, o, u\}, \quad P = \{x \in \mathbb{R} : x + 2 \leq 3\}$$

These are also sets:

$$\emptyset = \{\}, \quad Q = \{\emptyset\}, \quad B = \{1, 2, \{2\}, \{1, 2\}\}$$

Note that Q is not the empty set, it contains the empty set as an element of it!

Activity 1: Well-Defined Collections

Decide whether each description defines a set.

1. The planets in the solar system.
2. The students in this classroom.
3. The beautiful paintings in the world.
4. The prime numbers less than 20.
5. The difficult problems in this course.
6. Real numbers whose square is -1.

Activity 2

Give an example of a set that has three elements, say x, y and z , such that x is an element of y and y is an element of z .

Activity 3

How many different sets can be formed by selecting elements out of a, b and c ?

Methods of Describing Sets

Roster Method

List the elements explicitly:

$$A = \{1, 2, 3, 4, 5\}.$$

Set-Builder Method

Describe the property that elements satisfy:

$$A = \{x \in \mathbb{N} : x < 6\}.$$

More specifically, give a predicate $p(x)$, whose domain of discourse is D , we can construct the set

$$P = \{x \in D : p(x)\} = \{x \in D : p(x) \text{ is true}\}.$$

Descriptive Method

Describe the set in words:

$A =$ the set of natural numbers less than 6.

Activity 4: Convert Between Forms

Write each set using roster form, set-builder form and descriptive form where possible.

1. Even integers between 1 and 10.
2. Multiples of 3 less than 20.
3. Vowels in the English alphabet.
4. Prime numbers less than 30.

2. Membership and Subsets

Membership and Subset Relations

If x is an element of A , we write

$$x \in A.$$

If every element of A is also an element of B , then A is a subset of B , written

$$A \subseteq B.$$

- Membership $\in$ relates an element to a set.
- Subset $\subseteq$ relates a set to another set.

Activity 5: Taking Care of Common Misconceptions

Let $A = \{1, 2, 3\}$. Determine whether each statement is true or false.

1. $2 \in A$
2. $\{2\} \in A$
3. $\{2\} \subseteq A$
4. $2 \subseteq A$
5. $\emptyset \subseteq A$
6. $A \subseteq A$

Caution! Not all collections are sets.

For example the collection of all sets is not a set.

Let Λ be the collection of all sets. Then consider the subset $B = \{X \in \Lambda : X \notin X\}$

In this case Λ is not well-defined. However, there are also collections that are well-defined, but not considered to be sets. Mathematicians call them “proper classes”. But, this discussion is not under the context of this course.

Equality of Sets

Definition: Equality of Sets

Two sets A and B are equal if they have exactly the same elements.

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

This gives a powerful method for proving set identities: prove both inclusions.

Universal Set and Empty Set

The **universal set** U is the set of all objects under discussion.

The **empty set** is the set with no elements:

$$\emptyset = \{\}.$$

3. Set Operations

Let A and B be subsets of a universal set U .

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

$$A^c = \{x \in U : x \notin A\}.$$

Example

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Then

$$A \cup B = \{1, 2, 3, 4, 5, 6\},$$

$$A \cap B = \{3, 4\},$$

$$A \setminus B = \{1, 2\}, \quad B \setminus A = \{5, 6\}.$$

Activity 6

Let $U = \{1, 2, \dots, 10\}$, $A = \{2, 4, 6, 8, 10\}$. Find:

$$A^c, \quad A \cup A^c, \quad A \cap A^c.$$

Activity 7

Let $U = \{1, 2, \dots, 12\}$, $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8, 10, 12\}$. Find:

$$A \cup B, \quad A \cap B, \quad A \setminus B, \quad B \setminus A, \quad A^c, \quad B^c.$$

Cartesian Products

For sets A and B , the Cartesian product is

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

Example

Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Then

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

Remark

- In general, $A \times B \neq B \times A$.
- For finite sets, $|A \times B| = |A||B|$. Here, $|A|$ denotes the number of elements in the set.

Activity 8

Write the set by listing all the elements and determine the cardinality of each Cartesian product.

1. $\{0, 1\} \times \{x, y\}$
2. $\{a, b, c\} \times \{1, 2\}$
3. $\{1, 2\} \times \{1, 2\}$

4. Venn Diagrams (Examples)

A Venn diagram is a diagram used to show relationships between sets.

Usually, each set is represented by a circle, and the universal set is represented by a rectangle. Points inside a circle belong to that set, while points outside the circle do not belong to it. When two circles overlap, the overlapping region represents the elements common to both sets.

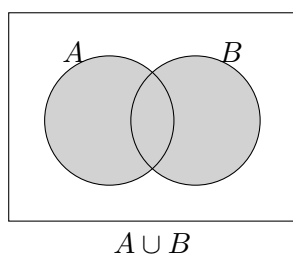
Venn diagrams are useful because they allow us to understand set operations visually. They are especially helpful when studying union, intersection, complements, and set identities.

Example

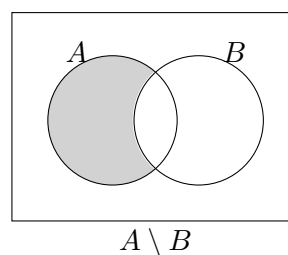
If A and B are two sets, then:

- the region inside both circles represents $A \cap B$,
- the region inside either circle represents $A \cup B$,
- the region outside B but in A represents $A \setminus B = A \cap B^c$
- the region outside both A and B represents $(A \cup B)^c$

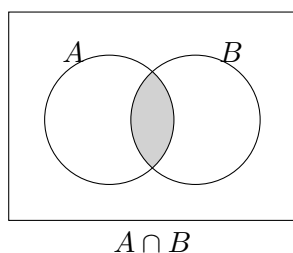
Venn Diagram: $A \cup B$



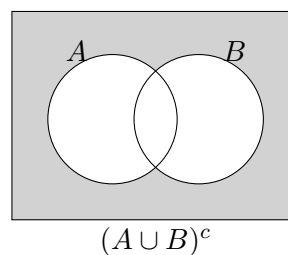
Venn Diagram: $A \setminus B$



Venn Diagram: $A \cap B$



Venn Diagram: $(A \cup B)^c$



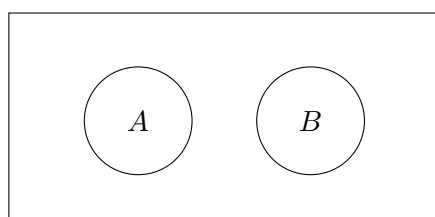
Special Venn Diagrams

In many problems, the relationship between two sets is not the general overlapping case. Some special cases are especially important.

Special Venn Diagram: Disjoint Sets

Two sets A and B are called **disjoint** if they have no common elements.

$$A \cap B = \emptyset.$$

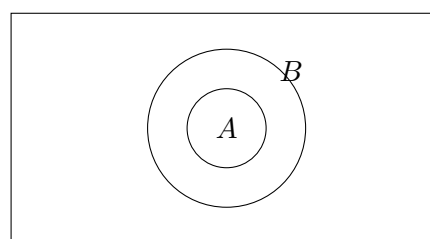


Disjoint sets: $A \cap B = \emptyset$

Special Venn Diagram: Subsets

If every element of A is also an element of B , then

$$A \subseteq B.$$



Subset situation: $A \subseteq B$

Example

Let $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$.

Then

$$A \cap B = \emptyset.$$

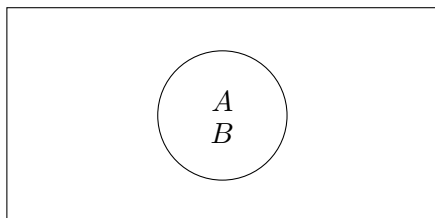
Consequences

If $A \subseteq B$, then

$$A \cup B = B \text{ and } A \cap B = A.$$

Special Venn Diagram: Equal Sets

If $A \subseteq B$ and $B \subseteq A$ then, $A = B$.



Equal sets: $A = B$

Important Point: This is why proving both inclusions is a standard method for proving set equality.

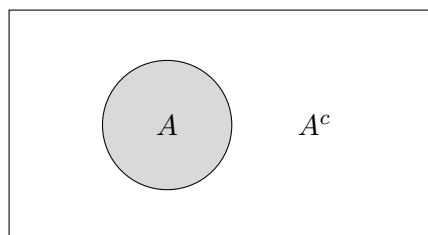
Special Venn Diagram: Complementary Sets

A set A and its complement A^c divide the universal set into two disjoint parts.

$$A \cap A^c = \emptyset,$$

and

$$A \cup A^c = U.$$

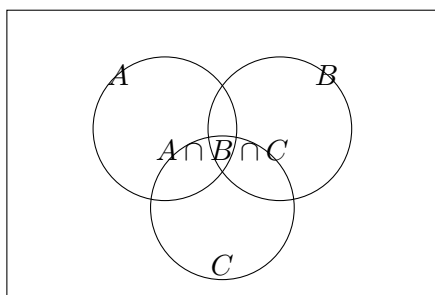


Complementary sets: $A \cup A^c = U$

Special Venn Diagram: Three Sets with Common Intersection

For three sets A , B , and C , the central region represents elements that belong to all three sets.

$$A \cap B \cap C.$$

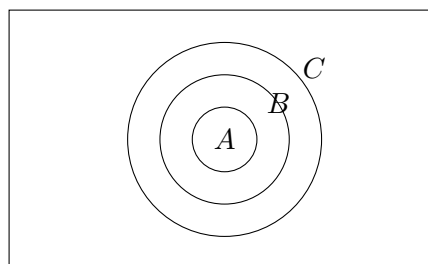


Three-set Venn diagram

Special Venn Diagram: Nested Sets

Sometimes several sets are contained one inside another.

$$A \subseteq B \subseteq C.$$



Nested sets: $A \subseteq B \subseteq C$

Consequences If $A \subseteq B \subseteq C$, then $A \subseteq C$.

Also, $A \cap C = A$, and $A \cup C = C$.

Activity 9: Interpret the Diagram

For each diagram above, write down:

1. A set identity suggested by the diagram.
2. An example of such sets.

5. Set Algebra

The following identities hold for all subsets $A, B, C \subseteq U$.

Commutative Laws

$$A \cup B = B \cup A, \quad A \cap B = B \cap A.$$

Associative Laws

$$(A \cup B) \cup C = A \cup (B \cup C), \\ (A \cap B) \cap C = A \cap (B \cap C).$$

Distributive Laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \\ A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Identity, Domination and Idempotent Laws

$$A \cup \emptyset = A, \quad A \cap U = A. \\ A \cup U = U, \quad A \cap \emptyset = \emptyset. \\ A \cup A = A, \quad A \cap A = A.$$

Complement Laws

$$A \cup A^c = U, \quad A \cap A^c = \emptyset.$$

De Morgan's Laws

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

6. Proving Set Identities

Proving Set Identities by Mutual Inclusion

To prove $X = Y$, prove $X \subseteq Y$ and $Y \subseteq X$.

Example : Distributive Law

For any three sets A, B and C ,

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Proof.

Let $x \in A \cap (B \cup C)$.

Then $x \in A$ and $x \in B \cup C$.

Therefore $x \in B$ or $x \in C$.

Case 1: If $x \in B$

Then $x \in A$ and $x \in B$

Thus $x \in A \cap B$.

$\therefore x \in (A \cap B) \cup (A \cap C)$

Case 2: If $x \in C$

Then $x \in A$ and $x \in C$

Thus $x \in A \cap C$

$\therefore x \in (A \cap B) \cup (A \cap C)$

So, in either case $x \in (A \cap B) \cup (A \cap C)$

Thus, for all x , $x \in A \cap (B \cup C) \implies x \in (A \cap B) \cup (A \cap C)$.

$\therefore A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Conversely, let $x \in (A \cap B) \cup (A \cap C)$.

Then $x \in A \cap B$ or $x \in A \cap C$.

Case 1: If $x \in A \cap B$

Then $x \in A$ and $x \in B$

Then $x \in A$ and $x \in B \cup C$

Thus $x \in A \cap (B \cup C)$.

Case 2: If $x \in A \cap C$

Then $x \in A$ and $x \in C$

Then $x \in A$ and $x \in B \cup C$

Thus $x \in A \cap (B \cup C)$

Hence in either case we have that $x \in A \cap (B \cup C)$

Thus, for all x , $x \in (A \cap B) \cup (A \cap C) \implies x \in A \cap (B \cup C)$.

$\therefore A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Hence, equality holds. □

Example : Difference and Union

For any three sets A, B and C ,

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

(The next proof is more compact and written in a paragraph, rather than organizing it in a line by line format. Also, try to notice certain logical equivalences used without mentioning in the proof.)

Proof. Let $x \in A \setminus (B \cup C)$. Then $x \in A$ and $x \notin B \cup C$. Hence $x \notin B$ and $x \notin C$. Therefore $x \in A \setminus B$ and $x \in A \setminus C$. Thus $x \in (A \setminus B) \cap (A \setminus C)$.

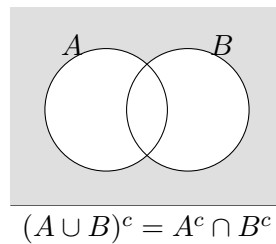
Conversely, let $x \in (A \setminus B) \cap (A \setminus C)$. Then $x \in A \setminus B$ and $x \in A \setminus C$. Hence $x \in A$, $x \notin B$, and $x \notin C$. Therefore $x \notin B \cup C$, so $x \in A \setminus (B \cup C)$. Hence equality holds. $\square$

Proving Set Identities Using Venn Diagrams

Example : De Morgan's Law

Prove using Venn diagrams:

$$(A \cup B)^c = A^c \cap B^c.$$

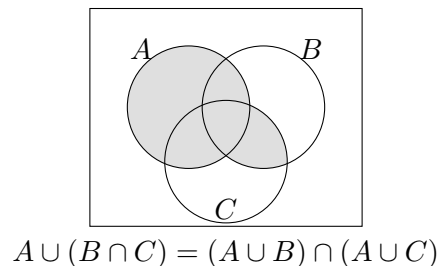


Both sides represent the region outside both A and B .

Example : Distributive Law

Verify using Venn diagrams:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

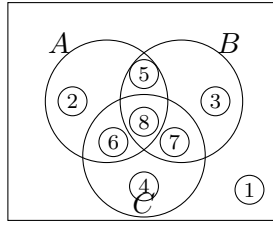


The shaded region is all of A , together with the overlap of B and C .

Example: Distributive Law

Let us prove the above identity $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ in another way.

We label the regions of the Venn diagram as follows.



The 8 regions of the Venn diagram

From the diagram, we can write down the areas that corresponds to each set:

$$A : \textcircled{2}, \textcircled{5}, \textcircled{6}, \textcircled{8},$$

$$B : \textcircled{3}, \textcircled{5}, \textcircled{7}, \textcircled{8},$$

and

$$C : \textcircled{4}, \textcircled{6}, \textcircled{7}, \textcircled{8}.$$

Now,

$$B \cap C = \textcircled{7}, \textcircled{8}.$$

Therefore,

$$A \cup (B \cap C) = \textcircled{2}, \textcircled{5}, \textcircled{6}, \textcircled{8}, \textcircled{7}, \textcircled{8}.$$

On the other hand,

$$A \cup B = \textcircled{2}, \textcircled{3}, \textcircled{5}, \textcircled{6}, \textcircled{7}, \textcircled{8},$$

and

$$A \cup C = \textcircled{2}, \textcircled{4}, \textcircled{5}, \textcircled{6}, \textcircled{7}, \textcircled{8}.$$

Therefore,

$$(A \cup B) \cap (A \cup C) = \textcircled{2}, \textcircled{5}, \textcircled{6}, \textcircled{7}, \textcircled{8}.$$

Since both sides give the same collection of regions,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

7. Exercises: Proving Identities Using Set Algebra

In the following problems, prove the identities using only the laws of set algebra. Do not use Venn diagrams or element arguments.

1. Easy Level: Let A and B be two subsets of the universal set U . Then:

(a) $A \cup (A \cap B) = A$

(d) $A \cap U = A$

(Hint: Use the distributive law.)

(e) $A \cup U = U$

(b) $A \cap (A \cup B) = A$

(f) $A \cap \emptyset = \emptyset$

(c) $A \cup \emptyset = A$

2. Intermediate Level: Let A and B be two sets. Then:

- | | |
|--|--|
| (a) $(A^c)^c = A$ | (d) $(A \cap B) \cup (A \cap B^c) = A$ |
| (b) $A \setminus B = A \cap B^c$ | (e) $(A \cup B) \cap (A \cup B^c) = A$ |
| (c) $A \setminus (A \setminus B) = A \cap B$ | (f) $(A \cap B) \cup (A^c \cap B) = B$ |

3. Advanced Level: Let A, B and C be three sets. Then:

- | | |
|---|---|
| (a) De Morgan's law:
$(A \cup B)^c = A^c \cap B^c$ | (d) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ |
| (b) $(A \cap B)^c = A^c \cup B^c$ | (e) $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$ |
| (c) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ | (f) $(A \cap B) \setminus C = (A \setminus C) \cap (B \setminus C)$ |

4. Challenge Level: Let A, B and C be three sets. Then:

- (a) Simplify $(A \cup B) \cap (A \cup B^c) \cap (A^c \cup B)$.
- (b) Prove that $(A \cup B) \cap (A^c \cup C) = (A \cap C) \cup (A^c \cap B)$.
- (c) Prove that $(A \cap B) \cup (A^c \cap C) \cup (B \cap C) = (A \cap B) \cup (A^c \cap C)$.
- (d) Show that $(A \cap B) \cup (A \cap B^c) \cup (A^c \cap B) = A \cup B$.
- (e) Let

$$X = (A \cup B) \cap (A \cup C) \cap (B \cup C).$$

Show that

$$X = (A \cap B) \cup (A \cap C) \cup (B \cap C).$$

(This is the set-theoretic analogue of a factorization identity.)

8. Power Sets

The power set of A , denoted $\mathcal{P}(A)$, is the set of all subsets of A .

If $A = \{1, 2\}$, then

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

Activity 10

- 1. Find $\mathcal{P}(\{a, b, c\})$.
- 2. Show that if $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.
Here $|A|$ is the number of elements in the set.

9. Cardinality

The cardinality of a finite set A , denoted $|A|$, is the number of elements of A .

Example

$$|\{1, 2, 3\}| = 3, \quad |\emptyset| = 0.$$

Activity 11

Find the cardinality of each of the following sets.

1. $\{1, 3, 5, 7, 9\}$

3. $\mathcal{P}(\{a, b, c, d\})$

2. $\{x \in \mathbb{N} : x \leq 20\}$

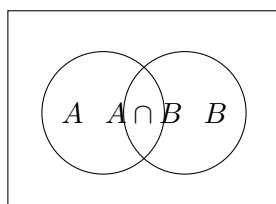
4. $\{0, 1\} \times \{a, b, c\}$

Inclusion–Exclusion Principle

Inclusion-Exclusion Principle

For finite sets A and B ,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$



$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example

In a class, 18 students study mathematics, 15 study computing, and 7 study both. Then

$$|M \cup C| = 18 + 15 - 7 = 26.$$

Inclusion-Exclusion Principle for Three Sets

For three finite sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

A Brief Discussion on Infinity

Some sets are infinite:

$$\mathbb{N} = \{1, 2, 3, \dots\}, \quad \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\},$$

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\}, \quad \mathbb{R} = \text{the set of all real numbers}.$$

Countably Infinite Sets

A set is **countably infinite** if its elements can be listed in a sequence.

The integers can be listed as

$$0, 1, -1, 2, -2, 3, -3, \dots$$

so $\mathbb{Z}$ is countably infinite.

The rational numbers $\mathbb{Q}$ are also countably infinite.

Uncountable Sets

The real numbers $\mathbb{R}$ are not countably infinite. They are **uncountable**.

Cantor showed that not all infinities have the same size.

Activity

Even more interesting is to observe that both $\mathbb{N}$ and $\mathbb{Z}$ have the same cardinality, even if $\mathbb{N}$ is a proper subset of $\mathbb{Z}$.

Discover the reason this is true.

Look up “Hilbert’s Hotel” to understand why infinity is so mind-boggling.

Summary

Set Algebra and Logic

Note the following analogy between set algebra and logic. Can you explain why such an analogy exists?

Logic	Set Theory
$P \wedge Q$	$A \cap B$
$P \vee Q$	$A \cup B$
$\neg P$	A^c
$P \Rightarrow Q$	$A \subseteq B$

For example,

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

corresponds to

$$(A \cup B)^c = A^c \cap B^c.$$

Review Questions

See whether you can answer the following:

1. Explain the difference between $x \in A$ and $x \subseteq A$.
2. Find $\{1, 2\} \times \{a, b, c\}$.
3. State De Morgan's Laws for sets.
4. If $|A| = 4$, what is $|\mathcal{P}(A)|$?
5. If $|A| = 20$, $|B| = 15$, and $|A \cap B| = 6$, find $|A \cup B|$.

Exercises (Ctd.)

5. Some easy questions.
 - (a) Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 6, 8\}$. Find $A \cup B$, $A \cap B$, $A \setminus B$, and $B \setminus A$.
 - (b) Let $U = \{1, 2, \dots, 10\}$ and $A = \{1, 3, 5, 7, 9\}$. Find A^c .
 - (c) Find $\{1, 2, 3\} \times \{a, b\}$.
 - (d) Find $\mathcal{P}(\{x, y, z\})$.
 - (e) Prove $A \cap A = A$.
 - (f) Prove $A \cup \emptyset = A$.

6. Proof Exercises: Prove

- | | |
|--|---|
| (a) $(A \cup B)^c = A^c \cap B^c$. | (d) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$. |
| (b) $(A \cap B)^c = A^c \cup B^c$. | (e) $(A \setminus B) \setminus C = A \setminus (B \cup C)$. |
| (c) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. | (f) $A \times (B \cup C) = (A \times B) \cup (A \times C)$. |

7. Challenge Problems

- (a) Prove $|\mathcal{P}(A)| = 2^{|A|}$ for finite sets.
- (b) Derive the three-set Inclusion–Exclusion formula.
- (c) Give an example showing that $A \times B \neq B \times A$.
- (d) Show that if $A \subseteq B$, then $B^c \subseteq A^c$.
- (e) Use both mutual inclusion and a Venn diagram to prove

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

8. Investigate:

- (a) Cantor and the birth of set theory.
- (b) Why $|\mathbb{N}| = |\mathbb{Z}|$.
- (c) Why $\mathbb{R}$ is uncountable.
- (d) Russell's paradox.
- (e) Cartesian products and coordinate systems.
- (f) Set algebra and digital circuits.
- (g) Set operations in databases.